

SRI CHARAN SRIPADI

AI Engineer · Agentic AI, LLM Evaluation & MLOps

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PROFESSIONAL SUMMARY

AI and Data Science MSc student (University of Liverpool) who builds and ships agentic AI systems end to end: multi-agent pipelines on Google ADK, fine-tuned and calibrated models, and the evaluation, serving and CI/CD around them. Recent open-source work targets open feature requests in Google's Agent Development Kit and AWS Strands Agents, with every result measured on held-out data and reported with its limitations. Proficient in Python, PyTorch and Hugging Face Transformers; contributor to the official Hugging Face Transformers repository (PR #46777). Eligible to work in the UK without sponsorship.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, C, C++

Agentic AI & LLMs: Google ADK, multi-agent systems, tool calling, Agent Skills (SKILL.md), LLM-as-judge, agent simulation and evaluation, RAG, LiteLLM, Groq, prompt engineering

ML & Fine-tuning: PyTorch, TensorFlow, Hugging Face Transformers, ModernBERT, scikit-learn, calibration (Platt, ECE), A/B testing (McNemar), clustering, computer vision (OpenCV, Dlib), NLP

MLOps & Serving: MLflow, Hugging Face Hub (model registry), FastAPI, Flask, REST APIs, Prometheus metrics, data versioning, Docker, GitHub Actions (CI/CD), GHCR, Google Colab GPUs

Cloud & Data: AWS (EC2, S3), Google Cloud Platform, Microsoft Azure, Vercel, MySQL, SQLite, FAISS, Chroma

Engineering: Pytest (unit, integration, offline model tests), Git, Linux, OOP, system design, Agile/Scrum

EDUCATION

MSc Advanced Artificial Intelligence and Data Science — University of Liverpool, UK | 2025 – Present

BTech in Artificial Intelligence and Data Science — KL University, India | 2021 – 2025 | GPA 9.4 / 10.0, First Class with Distinction

EXPERIENCE

Independent AI Engineer — Open-Source Agent Tooling & ML Systems | 2023 – Present

- Built and open-sourced two agent-infrastructure projects (SkillMiner, TinyJudge) that address open feature requests in Google ADK and AWS Strands Agents, from data generation and fine-tuning to Docker-based deployment.
- Designed and deployed end-to-end AI systems across computer vision, NLP and multimodal agentic pipelines; built Flask and FastAPI REST services with latency optimisation (<300 ms) and CI/CD via GitHub Actions.
- Wrote comprehensive Pytest suites (unit, integration, and offline tests with stand-in models) so pipelines run reproducibly in CI without API keys or model downloads.

Open Source Contributor — Hugging Face Transformers | 2025 – Present

- Submitted PR #46777 to the official Hugging Face Transformers repository, replacing wildcard imports with explicit named imports in the TimeSformer model module (resolves issue #45306). github.com/huggingface/transformers/pull/46777

PROJECTS

TinyJudge — Fine-tuned, Calibrated Judge for AI-Agent Behaviour | github.com/Charansripadi/tinyjudge

PyTorch · Hugging Face Transformers · ModernBERT · MLflow · FastAPI · Docker · GitHub Actions · HF Hub

- Fine-tuned ModernBERT-base (150M parameters) as an open alternative to LLM-as-judge for three agent checks: policy compliance, tool choice, and faithfulness of replies to tool results (AWS Strands Agents issue #4585).
- Generated a versioned dataset (SHA-256 manifest) of rule-labelled agent trajectories and LLM-written replies; on 208 held-out real LLM-agent conversations it reached **88.0% accuracy vs 81.2% for a 27B LLM judge** (Qwen 3.8).
- Per-task Platt calibration fitted on separate real data **halved calibration error** (ECE 0.20 to 0.10) and lifted faithfulness accuracy from 74% to 84%; a confidence cascade reached 89.9% while sending only 35% of cases to the LLM.
- MLOps pipeline: MLflow tracking, Colab T4 training (about 4 minutes), Hugging Face Hub as a versioned model registry, FastAPI service with Prometheus metrics and drift signals, and a Docker image that CI builds, smoke-tests and benchmarks behind an accuracy quality gate before publishing to GHCR.

SkillMiner — Multi-Agent System: Agent Traces to Validated Skills | github.com/Charansripadi/skillminer

Google ADK · Multi-agent · LiteLLM · Groq · scikit-learn · Pytest

- Built a multi-agent pipeline on Google ADK 2.10 that mines recurring workflows from agent session traces and writes them as Agent Skills (SKILL.md) that ADK loads natively, addressing Google ADK issue #5398.
- Created an ADK trace-recorder plugin, an LLM customer simulator with personas and hidden ground truth, and an LLM intent-induction miner: **purity 1.00 and ARI 0.93, vs 0.48 for a TF-IDF clustering baseline**.
- Added a Privacy Guard (PII redaction driven by tool-call values), a paired A/B Validator (exact McNemar) and a Drift Watcher; the mined refund skill learned "never refund an ineligible order" from violations seen in traces. Pilot A/B (8 pairs): procedure compliance 75% to 100%, policy violations 1 to 0, reported as not yet significant.
- Found and fixed an idempotency bug where retrying a rate-limited agent turn re-ran side-effecting tools, by moving retries to the model-call level; 23 offline tests.

FusionNet — Multimodal Agentic Video Analysis System | github.com/Charansripadi

Python · PyTorch · NLP · Agentic Pipelines · Docker · RAG

- Architected a multimodal agentic pipeline with 4 cooperating agents (Loader, Vision, Audio, NLP) coordinated via a FusionController, achieving <200 ms inter-agent latency.
- Generated structured JSON outputs (narrative summaries, semantic tags, emotion and energy estimates) with 91% label consistency across 50+ test videos; containerised with Docker and validated with Pytest integration tests.

Other projects: Driver Drowsiness Detection (real-time OpenCV/Dlib web app, 95% detection accuracy over 200+ sessions, 24 FPS Flask API, 35% latency reduction, Docker + GitHub Actions); Speech Emotion Recognition (7 classes, 88% accuracy on 10,000 samples, <300 ms inference, deployed on AWS EC2 with Docker).

CERTIFICATIONS

- TensorFlow Developer Certification — Google (2024)
- Google Cloud Associate Cloud Engineer (2024)